ARTICLE

Timing or tumour? The evidence behind glioma progression scores on post-treatment MRI

Sri Datta Ganesh Bandreddi[a] and Jessica Zhang[b]

[a]Department of Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA; [b]Department of Mechanical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA

ARTICLE HISTORY

Compiled November 4, 2026

ABSTRACT

Following treatment for aggressive glioma, changes on serial MRI must be interpreted alongside the treatment course. An automated progression score can rank surveillance scans, but cannot show whether it reflects MRI measurements or treatment context. We developed PRISM, a local multimodal decision support system that combines radiomic measurements and treatment context in an inspectable report. On MU-Glioma-Post, the development cohort, PRISM distinguished surveillance scans with recorded progression from other scans (AUC 0.82). Applied without refitting to external validation cohorts, prespecified models reached AUCs of 0.77 at LUMIERE and 0.80 at UCSF-ALPTDG. Across cohorts, the scores were linked to treatment timing at MU and enhancing volume at UCSF. PRISM presents the score with the measurements, treatment context, and provenance used to produce it. Across the three cohorts, an all 406 reports met predefined automatic checks against the fields in their structured records. Future work will evaluate the reports with clinicians.

KEYWORDS

glioma surveillance, external validation, clinical decision support

1. Introduction

After surgery and chemoradiotherapy for glioblastoma and other aggressive gliomas, serial MRI is read against earlier scans and the course of treatment. One early challenge is pseudoprogression. Treatment can produce new or increasing enhancement that resembles tumour growth. Pseudoprogression is concentrated within 12 weeks of completing chemoradiotherapy (Wen et al. 2023) and is more common in tumours with MGMT promoter methylation (Taal et al. 2008; Brandes et al. 2008).

Models intended to predict progression assign a score to each surveillance scan. On its own, a score gives an assessment without showing the information that informed it. It does not show whether the available information agrees, whether relevant information is missing, or how the model has performed in comparable clinical situations.

CONTACT Sri Datta Ganesh Bandreddi. Email: sbandred@andrew.cmu.edu; ORCID: 0009-0004-3617-6734

PRISM is a system that makes progression scores inspectable. For each scan, it keeps image measurements, treatment context, and separate model scores together in a record that is checked before it is written as a report. The name reflects this design. As a prism separates light, PRISM keeps image measurements and treatment context separate. Earlier systems have generated brain tumour reports from structured features extracted from segmentation (Heras Rivera et al. 2026; Valerio et al. 2025), and deterministic checks have compared generated clinical text with source records (Zeba et al. 2026). PRISM applies these ideas to glioma surveillance after treatment. Its language component does not interpret MRI or calculate a score. It writes the completed record, which an automatic checker compares with that record.

PRISM was developed on the public MU-Glioma-Post cohort and externally validated in the LUMIERE and UCSF-ALPTDG cohorts. The paper makes four contributions.

(1) It shows that progression scores can be supported by different information across cohorts. Treatment timing was associated with scores at MU, whereas enhancing volume was associated with scores at UCSF. AUC alone therefore does not establish what supports a score (Section 3.2).
(2) It introduces an evidence framework for each scan that presents image measurements, treatment context, and separate pathway scores together. It also records whether the pathways agree and how the model has performed in comparable situations, allowing the basis of a progression score to be inspected for an individual scan (Section 2.1).
(3) It provides external validation of models developed on the public MU-Glioma-Post cohort in the LUMIERE and UCSF-ALPTDG cohorts (Section 3.1).
(4) It provides a locally deployed reporting workflow grounded in the evidence record. PRISM generates a report only after the evidence record is complete and automatically checks the report against its source fields. Across the three cohorts, the generated reports met the predefined criteria for consistency with those records (Section 3.4).

The study evaluates model transport and whether reports remain faithful to their source records. Future work will evaluate the reports with clinicians.

2. Methods

2.1. PRISM

PRISM writes everything it computes for a scan, with the source of every value, into one structured record (Figure 1A). Table 1 names each score.

The timeline pathway scores a scan from clinical data and the patient's position in treatment, both taken as of the scan day. Where a cohort lacks treatment dates, a reduced timeline keeps the inputs it has (Table 1), with the flag for the first weeks after radiotherapy taken from the rater's note at LUMIERE and approximated at UCSF. The image pathway scores the scan from 27 measurements of the segmented tumour and its change (Table 1) and uses no dates or intervals. LUMIERE's masks have three compartments and no separate cavity, so there the image pathway uses the three volumes and their change since the previous scan, a reduction fixed before evaluation. The measurements come from the masks released with each cohort (Mahmoud et al. 2025; Suter et al. 2022; Fields et al.

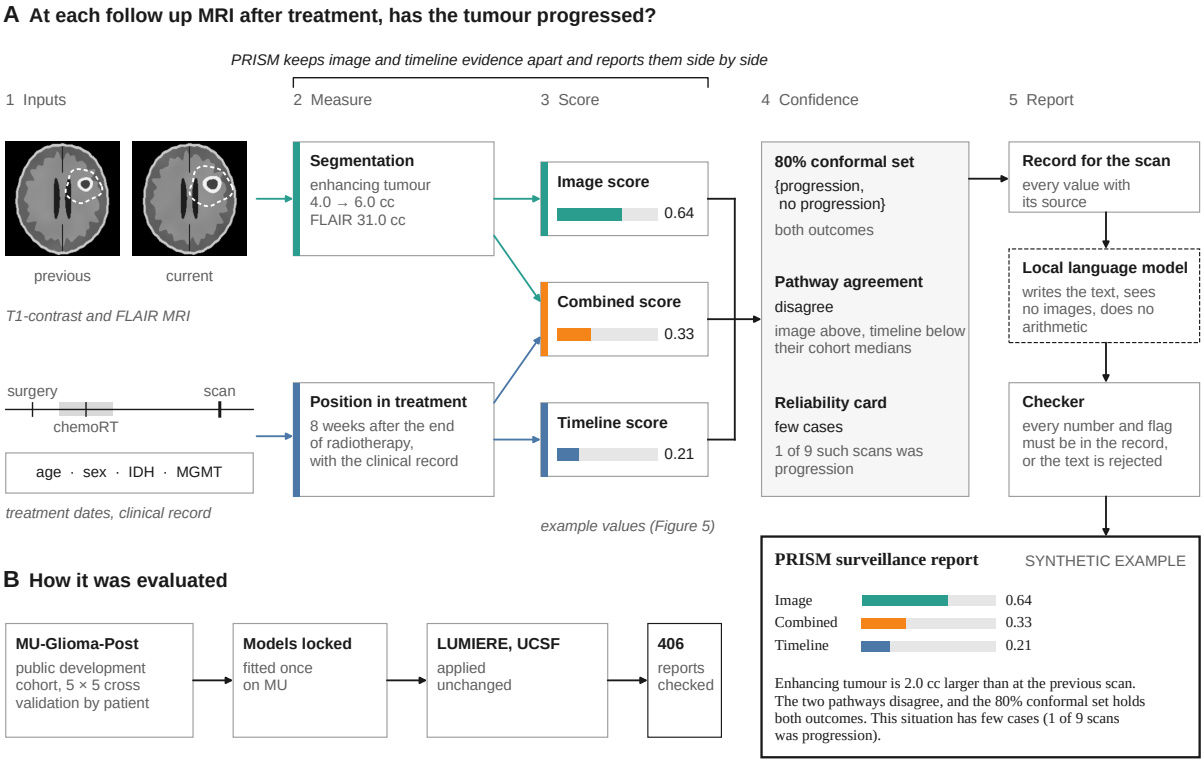


Figure 1. PRISM and how it was evaluated. (A) At each follow up MRI, PRISM keeps image and treatment evidence apart. The previous and current scans are segmented and the change in enhancing tumour and FLAIR abnormality is measured (image lane). The treatment dates and clinical record give the position in treatment (timeline lane). Each lane has its own score, and one model on both gives the combined score (values from the synthetic example of Figure 5). Three confidence statements are attached, every value enters one record with its source, and a local language model, given no images, writes the report, which a checker tests against the record. (B) Models were developed on MU by repeated cross validation, then fitted once and applied unchanged at LUMIERE and UCSF under rules fixed before any score was compared with an external label.

Table 1. The scores built from measurements and records. Every score listed is a ridge logistic regression except enhancing volume alone, which is not fitted. The others are fitted within the training data, with unknown IDH or MGMT status as the reference category and missing measurements imputed with an indicator. The networks tuned on the task are described in Section 2.3. Status indicates whether a model was planned before the analysis or not fixed in advance (†).

Score	Inputs	Dates	Status
Clinical data alone	Age, sex, IDH and MGMT status	No	Planned reference (LUMIERE †)
Timeline pathway	Clinical inputs; days since diagnosis and surgery; radiotherapy start, end and dose; chemotherapy; visit index	Yes	Planned
Reduced timeline	Clinical inputs; days since surgery; flag for the first weeks after radiotherapy	Partly	Planned at UCSF-ALPTDG; † at LUMIERE
Image pathway	27 measurements of the segmented tumour: volumes, shape and intensity, change since the previous scan, volumes relative to the patient's smallest	No	† on MU; planned at UCSF-ALPTDG and in the system tests
Images from one scan	The 19 measurements of the current scan	No	† on MU; planned at UCSF-ALPTDG
Combined score	Inputs of both pathways; externally, the reduced timeline, volumes and their change, and the interval since a previous scan	Yes; partly externally	† on MU (full form); planned and locked (reduced form) at LUMIERE and UCSF-ALPTDG
Planned imaging model	Timeline inputs, image measurements and three schedule inputs (interval since the previous scan, whether there was one, smallest volume since the end of radiotherapy)	Yes	Planned primary model on MU; not part of PRISM
Enhancing volume alone	Log enhancing tumour volume, no fitted model	No	† reference

2024).

The confidence has three parts. A conformal set split at the patient level at the 80% level (Angelopoulos and Bates 2023) holds one outcome when the score is clear and both when it is not. It comes from the combined model refitted on 70% of the MU patients and calibrated on the other 30% (within each training fold on MU) and is applied unchanged elsewhere, which assumes that external scans are exchangeable with MU's. The two pathways agree when both scores lie on the same side of their medians over the cohort's evaluation scans; because cohort medians need a batch of scans, we also tested medians fixed on MU, which can be applied to a single scan. The reliability card† (Mitchell et al. 2019) describes the scan's situation, its phase of care crossed with the change in enhancing volume, from that cohort's held out results. Because those results include the scan itself, the card describes it slightly optimistically. The card gives AUCs for a situation with at least 10 scans of each outcome (tested), counts only for one with fewer (few cases), and marks one with none as not tested.

A local language model (Qwen3.8-27B, temperature 0) writes the report from the record alone, so patient data do not leave the hospital. A deterministic checker rejects any number, flag or confidence statement that is not in the record, any missing required content, and any wording that offers a diagnosis, advice or a probability (Supplementary Section S19).

No public cohort releases the clinicians' reports for these scans, so we compared the report's content with a published standard instead, the follow up template of the Brain Tumor Reporting and Data System (BT-RADS) (Weinberg et al. 2018). Referring clinicians found reports written with this template less ambiguous than free prose (Gore et al. 2019), and the template reduced hedging (Zhang et al. 2020). Like the template, the report gives the position in treatment, the comparison with the previous scan and the change in enhancing tumour and FLAIR abnormality, here as mea-

sured volumes. It leaves out the template's category and management recommendation, because PRISM offers no diagnosis or advice, and it leaves out steroid and bevacizumab use, which the cohorts do not record (Supplementary Section S24).

2.2. Cohorts and the task

Grade 4 glioma is the primary population in all three cohorts (Table 2). We fitted each model once on MU and applied it unchanged, or locked, to the others. The task is to decide whether each scan is the one at which the cohort documented progression. On MU a positive is the first scan on or after the documented progression date, and a negative is a scan followed by at least 120 days without progression. At LUMIERE a positive is the first scan rated as progressive disease, and a negative is an earlier scan rated stable or responding. UCSF holds one pair of consecutive scans per patient, and we evaluated the second. A positive is a second scan whose progression date lies within 30 days of it and closer to it than to the first scan, and a negative is followed by at least 120 days without progression. On MU the intended use is scans from the start of radiotherapy, divided into three phases of care during radiotherapy, early after it (up to 12 weeks after its end), or later. MU results are for the intended use unless stated. Predicting progression within 120 days of a scan is a secondary task.

2.3. Evaluation

The locked models were fitted on all 181 grade 4 detection scans of MU, from every phase of care, and the external tests followed plans committed before any score was compared with an external label. The plans named all MU detection scans, but we fitted on the grade 4 scans. Fitting on all 229 scans of gliomas of every grade changed the LUMIERE AUC by less than 0.01. On MU we evaluated every model by five repeats of fivefold cross validation grouped by patient and report the mean pooled held out AUC with a 95% confidence interval (CI) from 1,000 bootstrap draws at the patient level. External AUCs carry a 95% CI from the same patient bootstrap. The MU intervals resample held out predictions and do not include the variance from refitting. No patient contributed to more than one of fitting, calibration and testing. We used repeated splits because single splits are unstable in small cohorts (An et al. 2021; Varoquaux 2018). An earlier radiomics model on MU, for example, reached 0.80 on one test split but averaged 0.72 when selected and tested again over 500 splits of the same design (Supplementary Section S8 and Figure S2). As a reference, we also fitted each UCSF model on UCSF by cross validation. To test whether deep networks see more than the measurements, we fine tuned segmentation networks to score scans directly and compared them with the same networks trained on permuted labels. Table 3 lists the questions fixed in advance and their answers. Results of analyses not fixed in advance carry a dagger (†). The disagreement flag and its two tests were planned, but the median rule that defines disagreement was written down after the results had been seen, so every disagreement result carries a dagger.

Table 2. The three cohorts. What each cohort records, and the scans in the detection task. Grade 4 glioma is the primary population.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Role	Development (repeated cross validation)	External, locked	External, locked
Access	Public	Public	Data use agreement
Scans per patient	Up to 6 scans with a label over 120 days (median 2)	Up to 10 scans with a label over 120 days (median 2)	One consecutive pair
Treatment dates	Surgery, radiotherapy start and end, dose, chemotherapy	Days from surgery only; rater flag for scans within three months of radiotherapy	Days from surgery; radiotherapy start only
Documented progression (the label)	Documented date in the clinical record	Expert response rating of each scan	Progression date supplied with the dataset
Segmentation masks	Automatic masks released with the dataset	DeepBraTumIA (three labels)	Masks provided with the dataset
Patients[a]	126	61	140
Age, median	62.5 (IQR 53–70)	62 (IQR 53–70)	57.5
Male	66 (52%)	32 (52%)	81 (58%)
IDH wildtype / mutant / unknown	112 / 8 / 6	38 / 1 / 22	81 / 13 / 46
MGMT methylated / unmethylated / unknown	45 / 62 / 19	26 / 28 / 7	71 / 43 / 26
Detection task[b]: scans with / without progression (patients)	62 / 84 (80) in the intended use; 64 / 117 (92) over all scans	73 / 105 (79)	29 / 40 (69), second scan only
Phase of care in the detection set	Progression: 56 of 62 later after radiotherapy; no progression: 51 of 84 early after radiotherapy	Not recorded	65 of 69 later after radiotherapy[c]

[a] Patient characteristics describe the cohorts of the secondary task over 120 days at MU-Glioma-Post (grade 4) and LUMIERE, and all grade 4 patients at UCSF-ALPTDG. Age is at diagnosis at MU-Glioma-Post and at surgery at LUMIERE.

[b] Positives are the scan at which progression was documented; negatives are defined per cohort in Section 2.2. The LUMIERE detection set has more patients than the cohort over 120 days because it includes patients who contribute only a progression scan. At MU-Glioma-Post, the 146 scans of the intended use are the 181 grade 4 detection scans less the 35 taken before radiotherapy started, and 42 grade 4 progression scans of patients with no scan carrying a label over 120 days are excluded. Reports were evaluated on the 114 intended use scans of the 60 MU-Glioma-Post patients not used to develop the prompt, on 162 of the 178 LUMIERE scans and on 130 of the 140 grade 4 second scans at UCSF-ALPTDG, the rest belonging to development patients (Section 3.4).

[c] UCSF-ALPTDG gives only the first radiotherapy start; the phase assumes a six week course. Two scans were early after radiotherapy and two had no radiotherapy date.

Table 3. Tests fixed in advance. Each rule was committed to the repository before the analysis it governs. AUC differences with 95% CIs.

Question	Rule fixed in advance	Outcome
Detection on MU-Glioma-Post		
Do image measurements add to the timeline?	Planned model ≥ 0.75 (lower bound ≥ 0.65); gain ≥ 0.03, CI above 0	**No**: AUC 0.837 (0.763–0.904); gain +0.015 (−0.032 to +0.063)
Do deep networks tuned on these data add?	Gain ≥ 0.03, CI above 0; control with permuted labels covers 0.5	**No**: no network gained; control valid, 0.483 (0.436–0.527)
External cohorts (LUMIERE, UCSF-ALPTDG)		
Combined score at LUMIERE	AUC ≥ 0.65	**Yes** (Section 3.1)
Image pathway at UCSF-ALPTDG	AUC ≥ 0.65, lower bound > 0.5	**Yes** (Section 3.1)
Locked image pathway adds to locked clinical data at UCSF-ALPTDG	Gain, CI above 0	**Yes**: +0.248 (+0.096 to +0.392)
Deep networks learn at UCSF-ALPTDG	Above the control with permuted labels; control covers 0.5	**Yes**: +0.201 (+0.094 to +0.306) over a control of 0.541 (0.480–0.603)
The system (MU-Glioma-Post, LUMIERE, UCSF-ALPTDG)		
Conformal coverage	≥ 0.78 at the 80% level in each cohort	**Yes** for the combined score, on point estimates; not for images at LUMIERE (0.708) or the timeline at UCSF-ALPTDG (0.696) (Section 3.3)
Does the combined score rank flagged scans less well?	AUC unflagged minus flagged, CI above 0	Disagreement[†]: **MU only**. Set holding both outcomes: **MU and LUMIERE**[a]
Does setting flagged scans aside beat matched removal?	Retained minus AUC after matched removal, CI above 0	Disagreement[†]: **MU only**. Set holding both outcomes: **UCSF only**[a]
Do reports pass the lexical check?	≥ 95% in each cohort	First evaluation **No** (every LUMIERE report failed); after one revision **Yes** (Section 3.4)
Is the tier stated where evidence is thin?	100%	**Yes** (Section 3.4)
Do reports follow a changed record?	≥ 27 of 30 perturbed records	**Yes** (Section 3.4)

Plans: detection (commit 2b4c927), UCSF-ALPTDG (0e7c9fe), the system (e7cba80), second report evaluation (4f7e7b9). Matched removal keeps the mix of care phases in the removed scans (random removal at LUMIERE, which has no phase). Over all scans of gliomas of all grades (a planned secondary population), the planned model gained +0.041 (+0.006 to +0.081) over the timeline, which may reflect its schedule inputs (Section 3.2).

[a] With each cohort's own medians, the rule the reports use; the median rule was written down after the results had been seen[†]. With medians fixed on MU, disagreement also met both tests at LUMIERE.

3. Results

3.1. *The timeline led at MU, and the combined and image models transferred without refitting*

In the intended use on MU (146 scans), the timeline pathway detected the progression scan with an AUC of 0.82 (95% CI 0.75–0.90) and the image pathway with 0.71 (0.63–0.78)†. Clinical data alone scored 0.53 (0.45–0.61; Figure 2). The combined score also reached 0.82 (0.74–0.89)†, with no detectable gain over the timeline (−0.001, −0.05 to +0.05). A planned model that added image measurements and three schedule inputs to the timeline also fell short of its target gain of 0.03 (Table 3). Within phases of care the timeline's gain was not detectable†, although its interval cannot exclude a gain as large as the overall one (Section 3.2). Forecasting progression within 120 days was harder, at 0.60 (0.52–0.69) in the intended use (Supplementary Section S23).

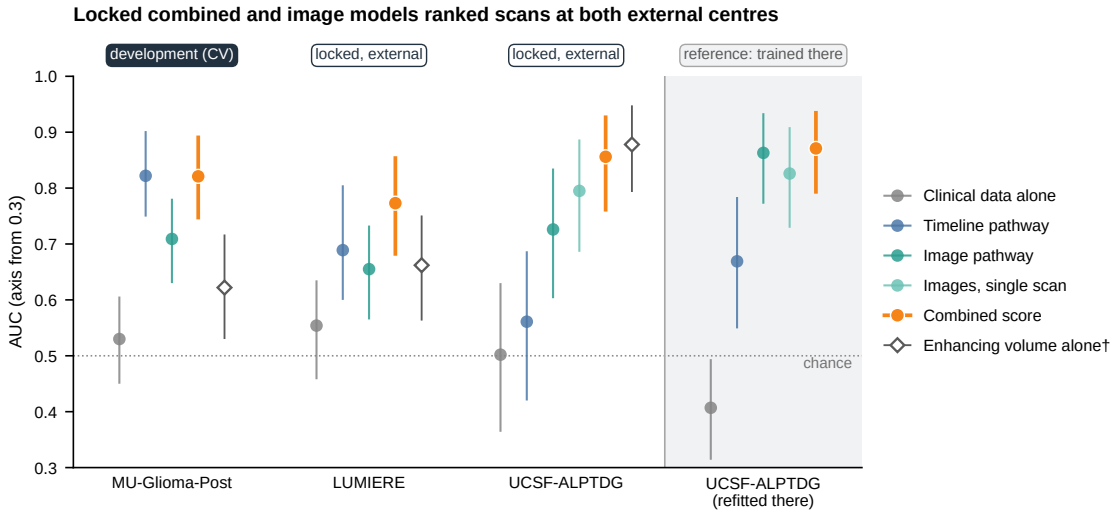


Figure 2. The combined and image models rank scans at two external centres without refitting. AUC for detecting the scan at which progression was documented. MU used repeated cross validation in the intended use. LUMIERE and UCSF used models fitted once on MU and applied unchanged, with the reduced timeline. The UCSF refit is a reference, where clinical data alone fall below 0.5, consistent with the downward bias of AUC estimated by cross validation without signal. At MU the combined score is its full form. Externally, it has a reduced form. At UCSF the images are shown for the model using one scan and for the image pathway. The image pathway and combined score on MU and the clinical data only, reduced timeline and image models at LUMIERE were not fixed in advance†. Open diamond: log enhancing tumour volume alone† (at LUMIERE, the 163 scans with enhancing tumour). Points have 95% CIs from bootstrap samples of patients on an axis from 0.3. The dotted line marks chance.

The locked combined model, in the reduced form used externally (planned, 0.80, 0.73–0.89, on MU by cross validation), reached 0.77 (0.68–0.86) at LUMIERE and met its planned transfer rule (Table 3). Its gain over the reduced timeline† was 0.08 (−0.01 to +0.16). At UCSF both planned image models transferred, the measurements from one scan reaching 0.80 (0.69–0.89) and the image pathway 0.73 (0.60–0.84); with coefficients fitted on MU, the image pathway scored 0.07 below the model using one scan (−0.13 to −0.01), and retrained at UCSF it gained 0.14 (0.03–0.26) over its locked version. The reduced timeline scored 0.56 (0.42–0.69) at UCSF, where its radiotherapy flag is approximated. The locked combined score reached 0.86 (0.76–0.93), but part of this reflects scan timing: the interval between the two scans was shorter before progression†.

As a probability, the combined score showed no detectable miscalibration in the large at LUMIERE† (observed over expected 0.97, 0.81–1.14) but overpredicted progression at UCSF (0.72, 0.55–0.91), and a threshold at the MU median† called 90% of UCSF scans positive.

3.2. The scores followed the treatment schedule at MU and enhancement at UCSF

At MU the timeline and combined scores depended mainly on where the patient was in treatment, and at UCSF the image scores followed the volume of enhancing tissue† (Figure 3). At MU most progression scans came more than 12 weeks after radiotherapy and most scans without progression within those 12 weeks (Table 2), so the schedule alone could separate most of them. An AUC within a phase pools only pairs of a progression scan and a scan without progression from the same phase of care; most such pairs come from the later phase (56 progression and 32 scans without progression). Adding the timeline to clinical data and images† changed the AUC by +0.11 (0.04–0.20) across phases of care and by −0.02 (−0.13 to +0.11) within them (Figure 3A). Within a phase, the image pathway scored 0.68 (0.58–0.77) and the timeline 0.64 (0.51–0.78), with no difference detectable.

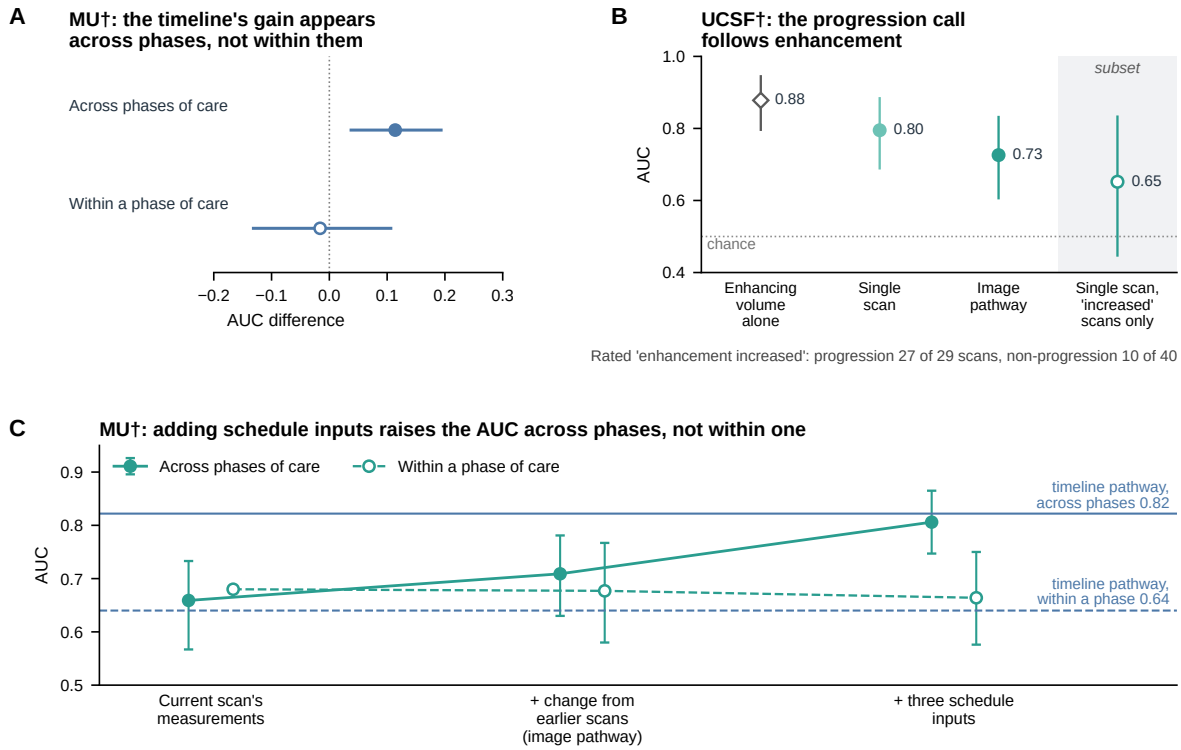


Figure 3. The scores followed the schedule at MU and enhancement at UCSF. (A) MU: gain from adding the timeline to clinical data and images, across and within phases of care. (B) UCSF: enhancing volume alone beside the locked image models; the open point is restricted to the 37 scans rated as showing increased enhancement. (C) MU: the image model built in three steps, across and within phases of care, against the timeline pathway; the third step adds the planned model's schedule inputs to the image inputs, without its timeline. The gain at the second step has an interval that includes zero, and the first point within a phase has no interval. The audits were not fixed in advance (†); the locked image models' AUCs over all 69 scans in (B) were planned. Points with 95% CIs from bootstrap samples of patients.

On MU the image model gained most from inputs that carry timing†. Adding change from earlier scans to the current scan's measurements, which gives the image pathway, raised the AUC by

0.05, with an interval that included zero (−0.003 to +0.106). Adding the planned model's three schedule inputs (Table 1) raised it by a further 0.10 (0.05–0.15) across phases of care but not within them (−0.01, −0.06 to +0.04) (Figure 3C).

At UCSF the progression labels coincided with a rated increase in enhancement†, so a model could score well there by tracking it. UCSF rated enhancement as increased at 27 of the 29 progression scans, and progression was dated a median of 7 days after the scan (Figure 3B). Enhancing volume alone reached 0.88 (0.79–0.95) there, above the point estimates of the locked image models, and above its AUC at MU and LUMIERE (Figure 2). Among the 37 scans rated as increased, the model using one scan reached 0.65 (0.44–0.84), so this sample cannot show whether the measurements add to that rating. Deep networks tuned on UCSF scans beat the control with permuted labels (Table 3), and their scores tracked enhancing volume†.

3.3. Coverage meets its target on point estimates and disagreement marks scans the combined score ranks less well at MU

The 80% conformal sets of the combined score met the planned coverage threshold of 78% on point estimates in all three cohorts, covering 78% to 87% of scans, and held a single outcome for 62% to 75% of scans (Figure 4A); the intervals at MU (76% to 88%) and UCSF (68% to 87%) reached below the threshold, and the image set at LUMIERE and the timeline set at UCSF fell below it. Coverage differed by outcome at the external centres†. LUMIERE covered 77% (67% to 87%) of progression scans and 94% of scans without progression, and UCSF 97% of progression and 65% (51% to 79%) of scans without progression. Calibrating the sets separately for each outcome on MU did not fix this, and coverage for scans without progression fell to 73% (63% to 82%) even on MU.

Disagreement between the pathways† marked scans that the combined score ranked less well at MU, but not at the external centres. On MU the pathways disagreed on 40% of scans. The combined score's AUC was 0.31 higher where they agreed (0.18–0.43), and setting the disagreeing scans aside gained 0.08 (0.05–0.12) over removing as many scans matched for phase of care, meeting both planned tests (Table 3; Figure 4B). With each cohort's own medians, the rule the reports use, neither test was met at LUMIERE or UCSF; with medians fixed on MU, both were met at LUMIERE but not at UCSF (Figure 4B). A conformal set holding both outcomes, used as a flag, met one of the two planned tests in each cohort (Table 3).

Only four situations on the reliability card† reached the tested tier (Figure 4C), one at MU and three at LUMIERE, where no phase of care is recorded and situations differ by enhancement only. No scan in these sets was in an untested situation. Every UCSF situation had few cases, and at MU no situation in the first 12 weeks after radiotherapy held more than three progression scans. In the planned descriptive comparison, tested situations had lower error than situations with few cases at LUMIERE (29% against 42% of scans misclassified at the cohort median) but not at MU (29% against 24%), where the scans with few cases are mostly clearly without progression early after radiotherapy.

3.4. After one revision, every report passed an automatic check against its record

Every report written by Qwen3.8-27B passed the lexical check against its record, 114 at MU, 162 at LUMIERE and 130 at UCSF. Of six other local models given the same prompt, which had been

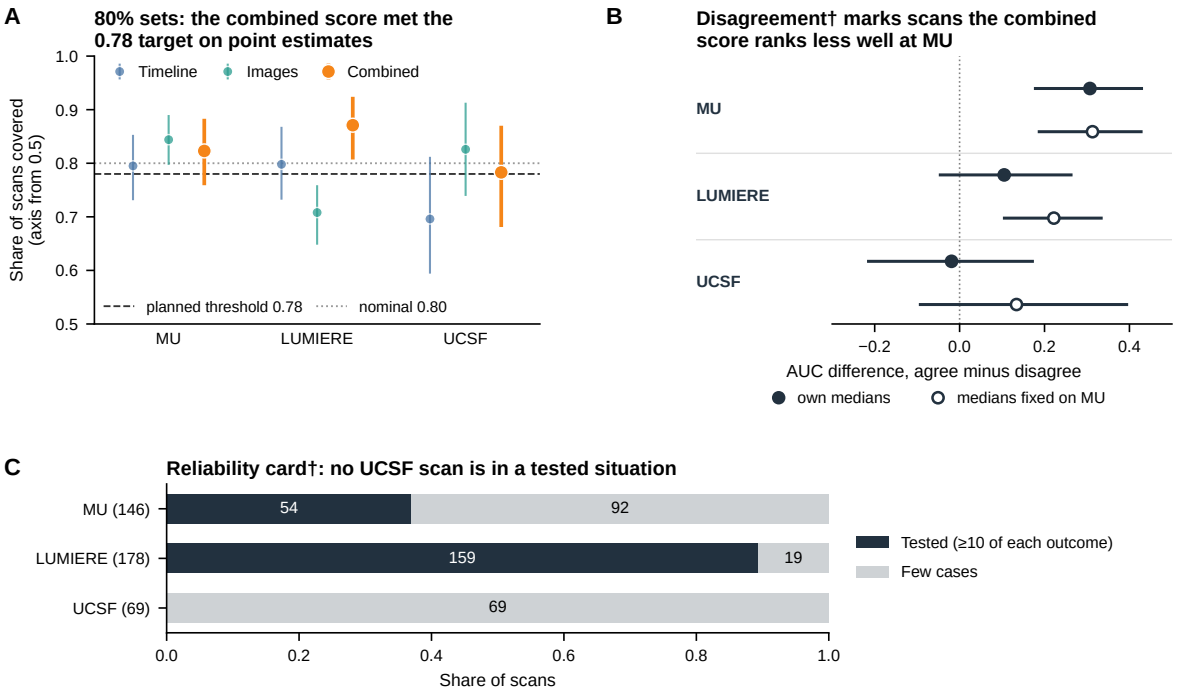


Figure 4. The combined score's conformal sets meet their coverage target on point estimates. Disagreement† marks scans where the combined score ranks less well at MU. (A) Coverage of the 80% conformal sets by cohort and pathway. The dashed line marks the planned threshold (0.78), and the dotted line marks the nominal level (0.80). (B) The combined score's AUC where the pathways agree minus where they disagree, with each cohort's own medians (filled) and with medians fixed on MU (open, using the other folds' medians on MU). The tests were planned, but the median rules were not†. (C) Scans whose situation reached the reliability card's† tested tier. The rest had few cases, and none was untested.

revised for Qwen3.8-27B, only Gemma-4 31B also did (Figure 5B). All 220 reports in situations with few cases or untested situations said so (71 at MU, 19 at LUMIERE and 130 at UCSF). When one confidence or measurement field was changed in 30 records from the three cohorts, 28 reports followed the change, against a planned minimum of 27 (Table 3). These results followed one revision of the prompt†. Before it, every LUMIERE report failed because a fixed sentence quoted a number held in no field, so we corrected the prompt, left the checker unchanged and evaluated again on patients not used for the revision. The check confirms that a report matches its record. It cannot judge emphasis, what a clinician would infer, or whether the segmentation is right.

4. Discussion

PRISM's combined and image models ranked scans at two other centres without refitting, but the evidence behind a score depended on how each cohort was scanned and labelled. At MU, the score mostly followed the treatment schedule and at UCSF it followed the enhancement on which progression was called†. An AUC cannot show which evidence a model used. The audits of Section 3.2 suggested it, and the separate pathways show the clinician which kind of evidence drives each score. Process variables can carry much of the apparent skill of an image model (Badgeley et al. 2019), and image models for glioblastoma surveillance have been benchmarked separately at the first and second scans after radiotherapy (Guo and Mirzaei 2025); we found no earlier study that compared image and timeline evidence within phases of care, or showed that what carries the same locked score differs between centres tested externally. The risk scale did not transfer to UCSF, so a new centre would recalibrate the score and set any threshold on local scans.

Because its modules are separate, PRISM runs on what a cohort holds, given scans and masks. The image pathway needs only those, which most public datasets have, and a version using one scan serves where there is no earlier scan. The timeline is added where dates exist. The reliability card is a lookup table, so a new cohort adds rows to it and a situation's tier can change as evidence accumulates.

Consider a scan eight weeks after radiotherapy in which the enhancing tumour has grown (Figure 5A). The timeline pathway scores such a scan low, because most scans at this point in treatment are not progression, and the image pathway scores it high, because enhancement has grown. The report shows both, states that they disagree and that the conformal set holds both outcomes, and records from the card that the situation has few cases (at MU, 1 of 9 such scans was progression†). The evidence PRISM holds for this situation is therefore too thin to support a call, in the weeks when pseudoprogression is most common, and the clinician decides with information it does not use, such as symptoms, steroid dose and the radiotherapy plan.

Each component of PRISM has been described before: longitudinal segmentation with measurement of change (Rudie et al. 2022), late fusion of separately trained image and clinical models (Boehm et al. 2022), fusion of clinical and imaging data to separate pseudoprogression from progression (Gomaa et al. 2025), conformal prediction for glioma segmentation (Elyassirad et al. 2025) and for triage at a deployment site (Angelopoulos et al. 2024), disagreement between modalities as uncertainty (Mahapatra et al. 2026), reliability from performance on similar cases (Habas et al. 2007), model cards (Mitchell et al. 2019), and reports written by local language models from brain tumour measurements (Heras Rivera et al. 2026; Valerio et al. 2025). The nearest systems combine some

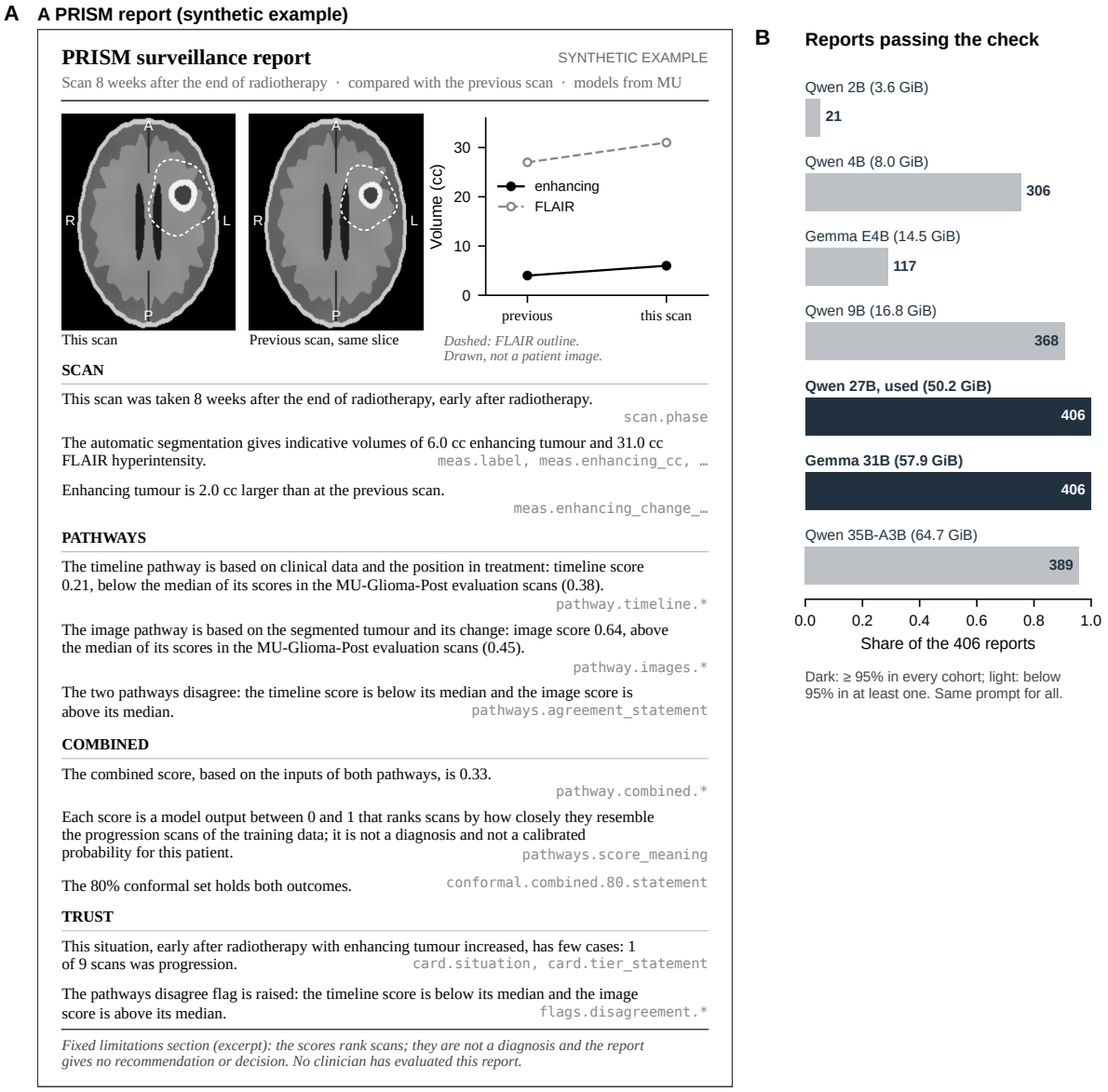


Figure 5. What the clinician receives, and the check. (A) A report for a synthetic scan eight weeks after radiotherapy with growing enhancement (Section 4). The segmentation figure is drawn here rather than taken from a patient, and the narrative is synthetic. Volumes and scores are invented, and the card row is MU's. Each sentence lists, in grey, the record fields it cites. The delivered report also holds tables of every value and a fixed limitations section (excerpt shown). (B) Reports passing the check, of 406, for seven local language models given the same prompt (Qwen3.5 2B, 4B and 9B, Gemma-4 E4B and 31B, Qwen3.8-27B used in PRISM, and Qwen3.5-35B-A3B), after one revision of the prompt† (Section 3.4). Dark bars show models for which at least 95% passed in every cohort, and the model weight size is in brackets.

of these. One assigns a BT-RADS category at follow up category from tumour volumes and treatment variables extracted by a language model, at one centre and without an uncertainty measure for each scan (Jabal et al. 2026); another keeps a ledger of evidence with a confidence for each claim in longitudinal neuro oncology, but works from reports rather than images and checks its evidence with a language model (Liu et al. 2026). On these cohorts, earlier work predicted a label for each patient on MU (Christodoulou et al. 2026) and response categories on LUMIERE ([cite Tikhonov 2025, arXiv:2509.06511; full reference not yet verified]), and a model trained on MU was applied unchanged at UCSF to predict survival (Ding et al. 2026); separately, Chen et al. (2026) predicted recurrence risk from preoperative and postoperative MRI across centres. None reports image and timeline evidence separately at each scan, or tests a progression call for each scan locked at two external centres. What PRISM adds is therefore not a new architecture but a combination: image and timeline evidence shown at each scan with a confidence from measured performance, tested at two external centres under rules fixed in advance.

PRISM was developed at one centre, and the two external tests are its only evidence that the ranking transfers. The label is progression as each centre documented it; no scan was read again, and at UCSF most progression scans were also rated as showing increased enhancement. Of the confidence measures, only average coverage met its planned threshold at all three centres, and only on point estimates. Disagreement† marked scans the combined score ranked less well at MU alone; that rule needs a batch of a cohort's scans, and the version using one scan (MU medians) was added after internal review. No UCSF scan fell in a tested situation. Subgroups are small, the analyses marked † need testing on new data, and no clinician has yet read the reports. PRISM forecast progression within 120 days less well than it detected progression at the scan (AUC 0.60, 0.52–0.69, in the intended use).

The next steps are to pool the rare situations (early progression after radiotherapy, and growing enhancement without progression) across cohorts, including the Burdenko and UCSD-PTGBM datasets, and test models tuned on them there; and a reader study to show whether separate evidence and a stated confidence change clinicians' decisions.

Conclusion

PRISM keeps evidence from images and treatment timelines apart and shows the clinician both scores at every scan with how far each has been tested there. Its combined and image models ranked scans at two other centres without refitting, while the scores followed the treatment schedule at MU and enhancement at UCSF, a difference the separate pathways make visible.

Acknowledgements

[PENDING AUTHOR DECISION].

Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other models with open weights in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no data at the patient level data through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

Funding

No funding was obtained for this work.

Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of datasets with identifiers removed (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test; their commits are listed in Table 3. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

References

An C et al. 2021. Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results. PLOS ONE. 16(8):e0256152. https://doi.org/10.1371/journal.pone.0256152

Angelopoulos AN, Bates S. 2023. Conformal prediction: A gentle introduction. Found Trends Mach Learn. 16(4):494–591. https://doi.org/10.1561/2200000101

Angelopoulos AN et al. 2024. Conformal triage for medical imaging AI deployment. [Preprint, medRxiv]

Badgeley MA, Zech JR, Oakden-Rayner L et al. 2019. Deep learning predicts hip fracture using confounding patient and healthcare variables. npj Digit Med. 2:31. https://doi.org/10.1038/s41746-019-0105-1

Boehm KM, Aherne EA, Ellenson L, Nikolovski I et al. 2022. Multimodal data integration using machine learning improves risk stratification of high-grade serous ovarian cancer. Nat Cancer. 3:723–733. https://doi.org/10.1038/s43018-022-00388-9

Brandes AA, Franceschi E, Tosoni A et al. 2008. MGMT promoter methylation status can predict the incidence and outcome of pseudoprogression after concomitant radiochemotherapy in newly diagnosed glioblastoma patients. J Clin Oncol. 26(13):2192–2197. https://doi.org/10.1200/JCO.2007.14.8163

Chen C, Guo F, Huang D et al. 2026. Predicting recurrence risk of glioblastoma based on preoperative-postoperative longitudinal MRI: A multicenter study. Bioengineering. 13(6):668. https://doi.org/10.3390/bioengineering13060668

Christodoulou RC, Vamvouras G, Pitsillos R, Solomou EE, Georgiou MF. 2026. Explainable radiomics with probability calibration for postoperative glioblastoma surveillance. Eur J Radiol Artif Intell. 5:100074. https://doi.org/10.1016/j.ejrai.2026.100074

Ding T, Zou Y, Chen C, Shah M, Tian Y. 2026. CLARITY: Medical world model for guiding treatment decisions by simulating context-aware disease trajectories. [Preprint, arXiv:2512.08029]. https://arxiv.org/abs/2512.08029

Elyassirad D, Gheiji B, Vatanparast M, Ahmadzadeh AM, Faghani S. 2025. CONSeg: Voxelwise uncertainty quantification for glioma segmentation using conformal prediction. Am J Neuroradiol. 46(12):2553–2560. https://doi.org/10.3174/ajnr.A8914

Fields BKK et al. 2024. The University of California San Francisco adult longitudinal post-treatment diffuse glioma MRI dataset. Radiol Artif Intell. 6(4):e230182. https://doi.org/10.1148/ryai.230182

Gomaa A, Huang Y, Stephan P et al. 2025. A self-supervised multimodal deep learning approach to differentiate post-radiotherapy progression from pseudoprogression in glioblastoma. Sci Reports. 15:17133. https://doi.org/10.1038/s41598-025-02026-7

Gore A et al. 2019. Institutional implementation of a structured reporting system: Our experience with the Brain Tumor Reporting and Data System. Acad Radiol. 26(7):974–980. https://doi.org/10.1016/j.acra.2018.12.023

Guo W, Mirzaei G. 2025. Timepoint-specific benchmarking of deep learning models for glioblastoma follow-up MRI. Cancers. 18(1):36. https://doi.org/10.3390/cancers18010036

Habas PA, Zurada JM, Elmaghraby AS, Tourassi GD. 2007. Reliability analysis framework for computer-assisted medical decision systems. Med Phys. 34(2):763–772. https://doi.org/10.1118/1.2432409

Heras Rivera JE et al. 2026. BTReport: A framework for brain tumor radiology report generation with clinically relevant features. [Preprint, arXiv:2602.16006]. https://arxiv.org/abs/2602.16006

Jabal MS et al. 2026. Agentic automation of BT-RADS scoring: End-to-end multi-agent system for standardized brain tumor follow-up assessment. [Preprint, arXiv:2603.21494]. https://arxiv.org/abs/2603.21494

Liu Y et al. 2026. TumorBoard: Evidence-grounded multi-agent decision support for longitudinal neuro-oncology. [Preprint, arXiv:2608.03190]. https://arxiv.org/abs/2608.03190

Mahapatra D et al. 2026. EnTrust: Modeling inter-modal conflict for trustworthy multimodal medical image analysis. [Preprint, arXiv:2606.21384]. https://arxiv.org/abs/2606.21384

Mahmoud E, Gass J, Dhemesh Y et al. 2025. MU-Glioma Post: A comprehensive dataset of automated MR

multi-sequence segmentation and clinical features. Sci Data. 12(1):1847. https://doi.org/10.1038/s41597-025-06011-7

Mitchell M, Wu S, Zaldivar A et al. 2019. Model cards for model reporting. In: Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19). ACM. p 220–229

Rudie JD, Calabrese E, Saluja R et al. 2022. Longitudinal assessment of posttreatment diffuse glioma tissue volumes with three-dimensional convolutional neural networks. Radiol Artif Intell. 4(5):e210243. https://doi.org/10.1148/ryai.210243

Suter Y, Knecht U, Valenzuela W et al. 2022. The LUMIERE dataset: Longitudinal glioblastoma MRI with expert RANO evaluation. Sci Data. 9(1):768. https://doi.org/10.1038/s41597-022-01881-7

Taal W, Brandsma D, de Bruin HG et al. 2008. Incidence of early pseudo-progression in a cohort of malignant glioma patients treated with chemoirradiation with temozolomide. Cancer. 113(2):405–410. https://doi.org/10.1002/cncr.23562

Valerio AG, Trufanova K, de Benedictis S, Vessio G, Castellano G. 2025. From segmentation to explanation: Generating textual reports from MRI with LLMs. Comput Methods Programs Biomed. 270:108922. https://doi.org/10.1016/j.cmpb.2025.108922

Varoquaux G. 2018. Cross-validation failure: Small sample sizes lead to large error bars. NeuroImage. 180:68–77. https://doi.org/10.1016/j.neuroimage.2017.06.061

Weinberg BD et al. 2018. Management-based structured reporting of posttreatment glioma response with the Brain Tumor Reporting and Data System. J Am Coll Radiol. 15(5):767–771. https://doi.org/10.1016/j.jacr.2018.01.022

Wen PY, van den Bent M, Youssef G et al. 2023. RANO 2.0: Update to the Response Assessment in Neuro-Oncology criteria for high- and low-grade gliomas in adults. J Clin Oncol. 41(33):5187–5199. https://doi.org/10.1200/JCO.23.01059

Zeba M, Al Mamun A, Tithee KJ, Sutradhar D et al. 2026. Mitigating hallucinations in healthcare LLMs with granular fact-checking and domain-specific adaptation. Expert Syst Appl. 329. Preprint arXiv:2512.16189.

Zhang JY et al. 2020. Quantitative improvement in brain tumor MRI through structured reporting (BT-RADS). Acad Radiol. 27(6):780–784. https://doi.org/10.1016/j.acra.2019.07.028